Parboni Chattopadhyay

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Education

University of California, Irvine — Irvine, CA
B.S. Informatics (Human-Computer Interaction Specialization), Minor in Computer Science — *Jun 2026*

Experience

Software Engineer and LLM Product Manager (Contract) — Jul 2026 – Present
Systems Intelligence Health Equity Research Group, North Jersey Management Solutions — *New York, NY*

- Built a Python Windows desktop app (Tkinter/ttk, PyMuPDF) that automates OCR on sensitive clinical PDFs by rendering pages as 300-DPI images, cutting review time from 1 hour to a few minutes per document.
- Met privacy requirements by integrating OpenAI's GPT-4o Responses API with Zero Data Retention and stateless requests.
- Secured API keys with Python keyring and Windows Credential Manager, plus privacy-preserving audit logging.
- Automated releases with a GitHub Actions CI/CD pipeline that builds Windows executables via PyInstaller.
- Gathered privacy and usability requirements from the research team, defined MVP scope, and shipped v1, owning the SDLC from design to testing.

Software Engineering Intern — Jun 2025 – Sep 2025
Numecent — *Irvine, CA*

- Engineered a C++ Win32 framework that runs PowerShell scripts across SYSTEM and USER security contexts in a single workflow, enabling enterprise deployments that need both admin and user setup.
- Ensured reliable execution via token validation and privilege-aware CreateProcessAsUserW/LoadUserProfileW calls.
- Built PowerShell/JSON workflows to export and restore Windows users and groups with idempotent imports.
- Cut firewall-rule deployment from 3.76s to 0.05s (RealVNC) and 8.03s to 0.04s (AnyDesk) with bulk PowerShell, 98%+ faster.

Software Engineering Grader — Sep 2024 – Mar 2026
University of California, Irvine — *Irvine, CA*

- Graded 200+ Python projects per quarter for Intro to Software Engineering, running unit tests and reviewing OOP design.

Research

Undergraduate Research Software Engineer — Jun 2024 – Present
AI/LLM Education Impact Research — *UC Irvine, Donald Bren School of ICS*

- Built Python/Pandas ETL pipelines to clean and analyze survey data on generative-AI adoption in software engineering education.
- Scaled analysis of 1,200+ responses with LLM-assisted qualitative coding refined from historical codebooks.
- Automated processing of 14 open-ended survey questions via data dictionaries into reproducible codeword datasets.
- Used one-hot encoding and Cramér's V to link AI usage to satisfaction and integrity concerns (ICCSE 2026 paper).

Undergraduate Research Software Engineer — Jan 2026 – Jun 2026
Pediatric Sleep Research Lab — *UC Irvine School of Medicine*

- Built a Unity/C# cognitive-assessment game from clinical requirements, winning 2nd Place nationally at Capstone Design.
- Refined age-appropriate interfaces with clinicians through iterative usability testing with 30+ users.

Systems Engineer — Aug 2025 – Dec 2025
NASA L'SPACE Research Program — *Remote*

- Contributed mission architecture, requirements, and subsystem design to a NASA proposal via trade studies.

Projects

Zot Spot | *Angular, TypeScript, Node.js, REST APIs* | devpost.com/software/zot-spot — Feb 2025

- Built a real-time menstrual-product availability platform for UC Irvine students with a six-person team, placing 6th at VenusHacks.
- Developed Angular/TypeScript views for location-based inventory, integrated with Node.js low-stock alerts.

HealthHarmonies | *HTML/CSS, JavaScript, Figma* | parbonichattopadhyay.github.io/HealthHarmonies — Mar 2026

- Developed a mobile-first healthcare resource app improving health-information access for underserved users.
- Designed responsive interfaces using HCI principles and Figma prototyping, meeting WCAG 2.1 AA standards.

Awards & Publications

- **2nd Place (National)**, Capstone Design Conference, Texas A&M University | **3rd Place**, UC Irvine ICS Senior Design Expo
- **Co-author**, *Student Perceptions and Attitudes Toward AI in Software Engineering Education at an R1 University*, ICCSE 2026
- **Featured**, UCI ICS News (2026), for the pediatric sleep-assessment game

Technical Skills

Languages: Python, C++, C#, Java, JavaScript, TypeScript, SQL, HTML/CSS, PowerShell
AI & Data: OpenAI API (GPT-4o), LLMs, Prompt Engineering, OCR, Pandas, ETL, Statistical Analysis
Frameworks & Tools: Angular, Node.js, Unity, Tkinter, PyMuPDF, httpx, keyring, PyInstaller, Git, GitHub Actions, Figma
Systems & Security: Win32 API, Windows Security Tokens, Windows Credential Manager
Practices: SDLC, CI/CD, REST APIs, OOP, Code Review, Agile, Requirements Analysis, Usability Testing, WCAG